

Guanlin WU

Email: 120010048@link.cuhk.edu.cn | Tel: (+86) 18753542673

Personal Website Link: <https://guanlinwu123.github.io>

EDUCATION

Chinese University of Hong Kong, Shenzhen	2020.09 - Current
• Computer Science and Engineering • MGPA: 3.470/4.0 • Core Courses: <i>Data Structure, Computer Architecture, Operating System, Software Engineering, Parallel Computing, Design and Analysis of Algorithm, Optimization, Stochastic Process, Reinforcement Learning</i>	
UC Berkeley	2018.06-2018.08
Academic Talent Development Program	Berkeley, CA

RESEARCH PROJECTS

University of California, Los Angeles/UAV Path Planning and Dynamic Obstacle Avoidance	2023.06 – 2023.09
<i>Research Assistant</i>	<i>Online</i>
• Propose a new UAV system based on 2D-LiDAR that including the Map Construction (based on Hector SLAM), the Path Planner (using RRT) and the Dynamic Obstacle Handler (using D3QN). • Design algorithms for LiDAR detection (with noise decreasing and motion trajectory) and Reset in RL training. • Develop the new training strategy of RL model for Dynamic Obstacle Handler is directly trained in the ROS-Gazebo-PX4 simulator with UAV model “iris_rplidar”. • Publication: Guanlin Wu, Zhuokai Zhao, Yutao He. “RELAX: Reinforcement Learning Enabled 2D-LiDAR Autonomous System for Parsimonious UAVs”. Submitted.	

SENIOR/AGV Resource Pool	2023.02 – 2023.04
<i>JAVA Engineer</i>	<i>Beijing, China</i>
• Propose several technical trace to implement the AGV Resource Pool: KM Matching algorithm, Dijkstra with time window, Dynamic Merging. • Develop a simulator based on OpenTCS for the performance testing on various possible algorithms.	

East China Normal University/Virtual Machine Allocation	2022.05 – 2022.11
<i>Research Assistant</i>	<i>Shanghai, China</i>
• Improved Best Fit Algorithm slightly for specific type of DVBP problem by using an innovative way of constructing the Balanced Binary Tree. • Managed to use Approximation algorithm to prove the ratio between modified Best Fit Algorithm and Optimal solution:$\text{BFT}(L) < \frac{20}{9} \text{OPT}(L)$ • Found the Virtual Machine data about their type from VMagent and HUAWEI. • Publication: Lifeng Guo, Changhong Lu, and Guanlin Wu. “Approximation algorithms for a virtual machine allocation problem with finite types.” Information Processing Letters 180 (2023): 106339.	

Tianjin Port/LiDAR-Based Hatch Localization	2021.07 – 2022.04
<i>Data Analyst</i>	<i>Tianjin, China</i>
• Used laser to collect information about shipment and stockpile and simultaneously created a 3D model using Point Cloud. • Applied Point Cloud and MATLAB to complete the rasterization process, one of the three stages of accomplishing the hatch localization • After scanning a reference hatch with size 23.55*15.35m using LiDAR, the hatch localization algorithm was able to measure the hatch accurately with the same dimensions, satisfying the anti-collision requirement of error margins less than 0.5 meters. • Publication: Zeyi Jiang, Xuqing Liu, Mike Ma, Guanlin Wu, and Jay A. Farrell. "LiDAR-Based Hatch Localization." Remote Sensing 14, no. 20 (2022): 5069.	

COURSE PROJECTS

Operating System Course Projects	2022.09 – 2022.12
<i>Independent Projects</i>	<i>CUHK(SZ)</i>

- **Learned about Kernel and loading a module into the Kernel:** Created a kernel thread and forked a process to execute the test program.
- **Used multithread programming to accomplish a game called “Frog crosses river” and implemented a thread pool:** This game is a multithread game, which requires forking threads for its implementation. The thread pool is one that acts as a pool consisting of several threads, which will perform the job after a user's request.
- **Implemented the simulation of a mechanism of virtual memory via GPU's memory (CUDA Programming):** Both physical memory and secondary memory have been implemented using shared memory and global memory in CUDA. Implemented the memory replacement policy for page faults. Implemented the mapping between virtual address (VA) to the physical address (PA). Implemented a paging system with swapping between shared memory and global memory.
- **Implemented the simulation of a mechanism of file system management via GPU's memory (CUDA Programming):** Implemented the logical file system that manages the metadata information, which includes all of the file-system structure except the actual data. Implemented the free-space manager that tracks the unallocated blocks and provides them when needed. Implemented a file-control block (FCB) which contains the information about the file, including ownership, permissions, and location of the file contents.

Design and Analysis of Algorithm Course Project: Pandemic Control Algorithm

2022.09 – 2022.12

Group Leader

CUHK(SZ)

- Implemented both the subtree-heavy algorithm and degree-heavy algorithm and conclude that subtree heavy performs 25% better than the other on average.
- Used Approximation algorithm to prove the ratio between Subtree-heavy Algorithm and Optimal solution, which is

$$S_{greedy} \geq \frac{1}{2} S_{optimal}$$

Database System Course Project: E-commerce Web Shopping Platform

2022.02 – 2022.05

Group Leader

CUHK(SZ)

- Developed a general understanding of the online shop platform.
- Utilized Java, Vue, Maven, and MySQL through the project.
- Implemented the CRUD operations through front-end with corresponding changes to back-end.
- Cultivated leadership and the ability of distributing tasks to members of the group.

Computer Architecture Course Projects

2022.01 – 2022.05

Independent Projects

CUHK(SZ)

- **Implemented a C++ translator for MIPS Instructions set to machine code:** This translator will scan the input file twice and return a table and a vector filled with each line of instructions. After processing, it will write the translated machine code into a .txt file.
- **Implemented a C++ simulator for the memory and register of the computer when running a MIPS asm file:** This simulator will display the change in Stack segment, Data segment(Dynamic data & Static data) and Text segment while running a MIPS code.
- **Implemented an Arithmetic and Logic Unit(ALU) by using Verilog:** This program implements a simple CPU which supports simple instruction parsing, Register Value fetching, and ALU functions such as add, addi, etc.
- **Implemented a 5-stage pipelined CPU by using Verilog:** This program mainly contains 5 clock cycle to execute one instruction. In the first clock cycle, the CPU read one instruction in the instruction memory with the newly given pc address. In the second one, the processor will divide the instruction to different parts and decode the MIPS instruction with the registers and control unit. In the third one, ALU will handle arithmetic, logical, shifting and conditional branch instructions. In the fourth one, data will be fetched from or stored to the data memory by data transfer instructions. In the fifth one, data will be written back to registers if needed.

AWARD

- School of Data Science **Dean's List** in 2022-2023

Others

- Language: English (TOEFL 108; GRE 326+4), Chinese (native)
- Computer skills: C, C++, Java, Python, mysql, Matlab, Point Cloud, Verilog, CUDA programming
- Interests: Philosophy, RAP, Work out, Basketball